Unit 2 Project Brief

Aerospace Payload Support Structure Challenge

Built to Survive: Aerospace Structures, Materials & Testing

Challenge Statement: Design, build, and test a lightweight payload support or aerospace structure that safely supports a required load while minimizing unnecessary mass.

Project Overview

In this project, your team designs and tests a lightweight aerospace structure that supports a required payload while managing mass, stiffness, stability, and material choice. Your design decisions must be backed by force diagrams, calculations, material reasoning, and physical test data.

Mission Scenario

Every aerospace structure must balance strength, weight, reliability, and material limits. Launch towers, payload frames, drone supports, rover chassis, aircraft structures, and satellite brackets must carry loads without wasting mass. Your team will design and test a structure that proves it can survive its mission.

Design Requirements

- Select an approved structure type such as payload support tower, launch support frame, drone landing support, truss, test stand, or aerospace bracket/frame model.
- Create free body diagrams or force diagrams showing the major external forces on the structure.
- Use structural ideas such as tension, compression, bending, truss geometry, centroid, moment, stiffness, or load path.
- Choose approved materials and explain why they are appropriate based on properties such as strength, mass, stiffness, density, or manufacturability.
- Build a prototype and test it using a controlled load procedure.
- Evaluate performance using load held, mass, deflection, structural efficiency, failure mode, or safety factor evidence.

Constraints

- Structure must fit within the approved size limits set for the classroom test area.
- Structure must use only approved materials, adhesives, fasteners, and fabrication methods.
- Load testing must follow the assigned safety procedure and may only happen in the approved test zone.
- Design must include a clear load path from payload/load point to support/base.
- Final design must be defended using both analysis and physical test evidence.

Required Engineering Evidence

Evidence Type	What to include
Analysis	Force diagram, load path notes, moment/equilibrium/truss reasoning, or other assigned calculations.
Material Selection	Material properties, density/mass notes, and explanation of material choice.
Design Development	Concept sketches, decision matrix, and selected structure plan.

Testing	Load, mass, deflection, failure mode, and structural efficiency data.
Final Recommendation	Evidence-based conclusion about whether the structure met the mission requirements.

Project Checkpoints

1. Mission requirements identified
2. Force diagram and concept sketches complete
3. Material plan approved
4. Prototype built
5. Load test and analysis complete

Success Criteria

Category	Evidence of success
Structural Performance	Structure supports the required load safely and with clear stability.
Analysis & Calculations	Force, load path, and material reasoning support the design choices.
Testing Quality	Testing is repeatable, documented, and connected to the criteria.
Design Communication	Final review explains tradeoffs among strength, mass, stiffness, and failure risk.

Final Submission

Submit a complete design package that includes your notebook evidence, sketches or CAD evidence, prototype/build evidence, test data, calculations or analysis, and a final design review presentation. Your final recommendation should clearly explain what your design does, how well it performs, and what should be improved next.